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Original Research

DGCL: Distance-wise and Graph Contrastive Learning for medication recommendation

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ABSTRACT

Medicine recommendation aims to provide a combination of medicine based on the patient's electronic health record (EHR), which is an essential task in healthcare. Existing methods either base recommendations on EHRs or provide models with knowledge of drug–drug interactions (DDIs) to achieve DDI reduction. However, the former models the patient's health history but ignores undesirable DDIs, while the latter lacks mining of patient health records and gets low recommendation accuracy. Therefore, this study contributes to research on personalized medication recommendations that consider drug interaction effects and models the patient's past medical history. In this paper, the Distance-wise and Graph Contrastive Learning (DGCL) framework is proposed. Specifically, we develop a two-stage neural network module for clinical record learning. We propose the distance detection loss to model the difference between the output distribution of current cases and historical records. In the DDI recognition and control task, DGCL proposes a graph contrastive learning method to jointly train the DDI knowledge graph and the electronic record graph, thereby effectively controlling the level of DDI for recommended medications. By comparing the performance on the MIMIC-III dataset with several baselines, DGCL outperforms other models in terms of efficacy and safety.

1. Introduction

Medication recommendations [1,2] have attracted much attention in recent years. Early medication recommendation models are instance-based and use current hospital visits. However, certain drawbacks are associated with these methods. For patients with chronic conditions or a detailed past medical history, instance-based approaches do not fully consider the patient's progression. Therefore, longitudinal approaches, such as [3–5], have been developed to use the time-dependence of longitudinal patient histories to provide more personalized advice. However, these methods are limited by the following:

- **Medical history links**: It is difficult to investigate the importance of certain cases because the correlation between past cases and current visits cannot be distinguished. Existing recommendation systems use single hospitalization records to forecast medication lists [6]. These systems do not comprehensively model the patient's medication history to provide a comprehensive treatment plan. The GAMENet recommendation system [7] uses dynamic memory to describe temporal dependence. However, these methods only use different combinations of patient records to make decisions when recommending medication and do not model the association between cases.
- **Uncontrollable DDI model**: Potential interactions in the patient's medicine combination may lead to unknown drug interactions and additional medical costs [8]. Avoiding interactions by spreading out or spacing medications is a common management idea. However, external medical knowledge can circumvent interactions at the recommendation stage and give safer drug combinations [9]. Therefore, access to accurate and comprehensive knowledge of DDIs, particularly during medication recommendations, is critical for patients. Past studies either did not consider DDIs or use external medical knowledge. For example, while DMNC [10] considers the time-dependence of patient cases when recommending medications, it does not model drug interactions. Complex drug interactions may inhibit drug efficacy and, in severe cases, may even lead to adverse reactions in patients [11].

Fig. 1 shows the medication recommendation process for one patient. Ideally, both the patient's historical medication use and potential interactions in the medication should be considered in the medication regimen. A subset of medications are selected for analysis from the patient's three visits, and the medications are represented by Anatomical Therapeutic Chemical (ATC) codes [12]. The patient takes medication

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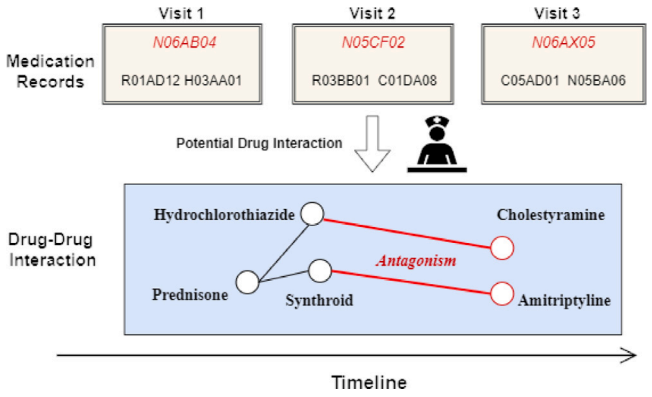


Fig. 1. Example of potential problems with medication recommendations.

for depression as indicated by the red drug code in the record in Fig. 1. The red medication at the first and third visits was an antidepressant, while the red medication at the second visit was for a severe insomnia problem. In addition, the patient had angina during the last two visits and has been taking a new medication for their heart since then. Thus, the patient's three visits are interrelated, there is a rich dependency between medications, and a rational medication recommendation system should consider the patient's past and present visits, and learn from their medication records in the EHR.

Another concern is DDIs in recommendations, and we have listed five drugs for analysis in Fig. 1: hydrochlorothiazide, amitriptyline, threonine, prednisone, and cholinesterase. In particular, cholinesterase and hydrochlorothiazide have an antagonistic effect. The former reduces the absorption of hydrochlorothiazide in the stomach and intestines, and should be taken at staggered times. In the case of Synthroid and amitriptyline, concomitant use of these two classes of drugs may increase both drugs' therapeutic and toxic effects, and increase the risk of arrhythmias in patients. Therefore, patients should not use these two drugs together. There are no significant interactions between hydrochlorothiazide, prednisone, prednisone, and cyproheptadine; thus, these drugs can be used as safe drug combinations for patients.

For the above mentioned problems, we design a novel medication recommendation model called DGCL, which learns the most relevant treatment records in the patient's past clinical records to assist in the current medication recommendation task. In addition, DGCL cleverly designs a graph contrastive learning framework that learns typical DDI knowledge graphs using EHRs, markedly suppressing drug interactions in recommended drugs, as demonstrated in this study's experiments. The proposed DGCL makes the following contributions:

- We designed a two-stage training strategy to manage medical history associations. Specifically, we design a medical attention module to record patient history information. Then, we propose a distance neural network that measures the proximity of historical and current cases, and retains appropriate treatment to assist with recommendation tasks.
- We have designed a graph sampling contrastive learning scheme to effectively control the DDI rate by studying the internal correlation of EHR and DDI graphs. Experiments have highlighted the importance of graph contrast learning, effectively suppressing interactions between EHR graphs.
- We conduct extensive experiments on the MIMIC-III baseline dataset to validate and analyze the effectiveness of DGCL. Experimental results show that DGCL outperforms most baseline methods and can make practical medication recommendations at lower DDIs.

2. Related work

Existing medication recommendation algorithms can be grouped into three categories: **rule-based recommendation, instance-based recommendation** and **longitudinal recommendation**.

Rule-based methods rely on regulations to access the knowledge of medical experts. For example, [13] attempted to develop hard-coded protocol-based recommendations for adaptive treatments for adolescent depression. Lakkaraju et al. [14] study the probability distribution between patient information and treatment. In general, models that use fixed rules are less generalizable. Additionally, it is difficult to add traditional rules to a well-established knowledge base.

In instance-based recommendation methods, the leap [6] extracts feature information from current patient instances and uses a multi-label medication recommendation setting. Those authors draw on the idea of label prediction tasks that consider both patient information and DDIs. Wang et al. [15] use demographic, diagnostic and procedural information and fuse them using linear methods. Most of these methods focus on individual patient instances and do not adequately model past information.

Longitudinal approaches have been proposed to exploit the time dependence in clinical histories, and such recommendation systems aim to exploit the time dependence between patients' past visits. Bajor and Lasko [16] suggest using recurrent neural networks (RNNs) to predict medication treatments based on the sufferer's EMRs. Although the model will recommend specific medication for the patient, it cannot manage concurrent disease conditions and give medication combinations. Le et al. [10] introduced a dual memory neural network to manage long-range dependence for medication recommendation. However, this approach does not consider the relationships between medications; thus, Shang et al. [7](GAMENet) constructed a graph based on the co-occurrence of medications in EHRs and used GNN to obtain embeddings of these medications.

Despite their initial success, these methods, cannot effectively manage DDIs. Existing models either do not sufficiently use DDI knowledge or lack DDI constraints when recommending medications. This study proposes using external DDI knowledge to generate knowledge graphs and design a graph contrastive learning framework to constrain the DDI rate with recommended medications to effectively control potential drug interactions. GAMENet also introduces external DDI knowledge but only calculates a difference between the case graph and the DDI graph, and the neighborhood information between medications on the entire knowledge graph is lost, which cannot manage multidrug interactions. The proposed model retains both global drug neighborhood information and dynamically distinguishes drug-specific interaction neighborhoods to adequately constrain drug interactions for the recommended combinations.

3. Problem formulation

Definition: The $E_t = \{d^t, p^t, m^{t-1}\}$ can denote cases of the patient, where d^t, p^t and m^{t-1} represent diagnostics, procedures, and medication records, respectively. Thus, medication recommendation can be simplified to occur when we obtain cases and historical medication of the patient. Our goal is to train the DGCL model to recommend drugs m^t while reducing the interactions in drug combinations.

3.1. Electrical health records (EHR)

Patient EHR data are collected in medical code carriers (e.g., diagnoses, procedures, and medications). Formally, we model patient V as a continuous sequence $X_V = [x^1, x^2, \ldots, x^i]$, where i is the total number of visits for patient V. For the patient with i visits, their EHRs are encoded into the 64-dimensional space, where diagnosis, procedure and medication information are denoted as d, p, m. The ith visit is given by $x^i = [d^i, p^i, m^i]$, where $d^i \in \{0,1\}^D$, $p^i \in \{0,1\}^P$, $m^i \in \{0,1\}^M$ are multi-hot vectors that represent different visit information, and D, P, M are medical code sets.

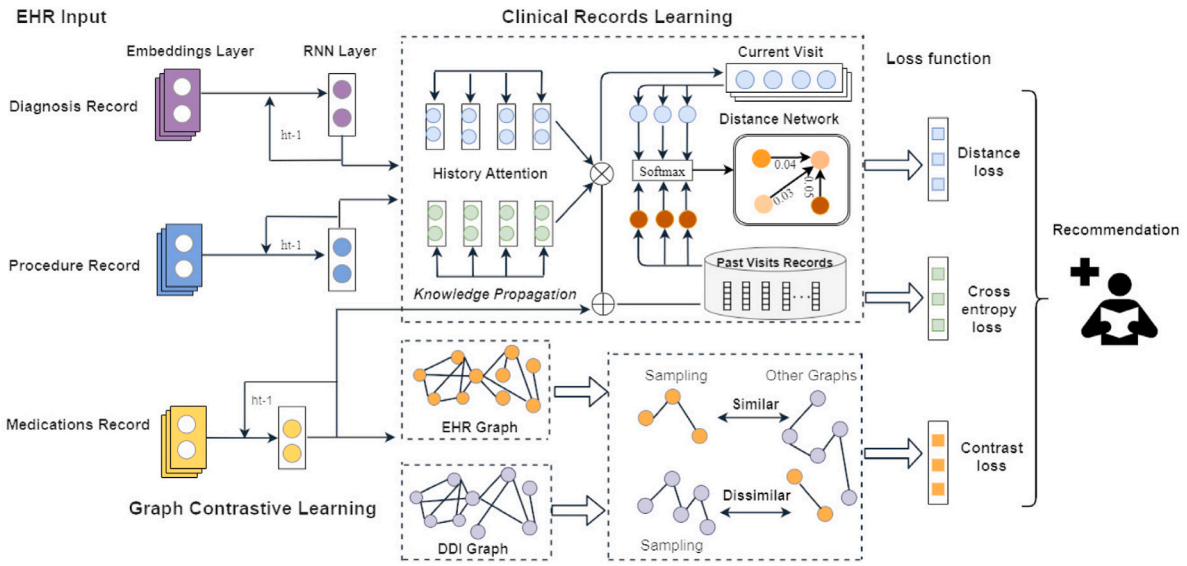


Fig. 2. Framework of the proposed DGCL.

3.2. EHR and DDI graph

The EHR graph and DDI graph can be formalized as $G_e = \{A, X_e\}$ and $G_d = \{A, X_d\}$, respectively. The node-set A represents the set of medications, X_e is the edge set of medication in EHR graphs and X_d is the edge set of DDIs graphs between a pair of medications. We define the adjacency matrix $E_e, E_d \in \mathbb{R}^{|M\times M|}$. Specifically, if the medicine i exists in the jth recommendation combinations, $E_e[i,j] = 1$. For E_d, only paired drug–drug interactions are considered, $E_d[i,j] = 1$ when the drug i interacts with the jth drug combination.

4. Methods

4.1. OVERVIEW

In this section, we review the various modules of the DGCL. The overview is shown in Fig. 2. The DGCL model includes four components: (1) **patient representation module,** which encodes patient cases from EHR data; (2) **clinical records learning module,** which is designed as a two-stage training. In the clinical records learning module, medical history attention is first designed to determine global information about the patient. Then, we use a distance neural network to constrain the probability distribution between current medication and patients' knowledge, and filter out similar medication records to assist in treatments (3) **graph contrastive learning framework,** where patient medication information and external DDIs knowledge are encoded into a graph structure separately, and graph sampling is performed for specific drugs to construct positive and negative samples of the drugs to be learned for contrastive learning to control the DDIs rate in the recommended drugs; (4) **training and inference,** drug recommendations must find an optimal balance between recommendation accuracy and safety. Therefore, we design a combination loss to obtain the final drug output.

4.2. EHR embedding

The patient's health status can be encoded by their diagnostic and procedural information. Therefore, we predict medication combinations based on patient EHRs. DGCL first embeds d^t, p^t and m^{t-1} into a continuous vector representation as Eq. (1) shows, where $W_d \in \mathbb{R}^{D\times d}$, $W_p \in \mathbb{R}^{P\times d}$ and $W_m \in \mathbb{R}^{M\times d}$ are embedding matrices to learn:

$$\begin{aligned} x_d^t &= W_d d^t \\ x_p^t &= W_p p^t \\ x_m^{t-1} &= W_m m^{t-1} \end{aligned} \tag{1}$$

4.2.1. Patient representation

To implement personalized medication recommendations that are tailored to patient EHR data, we design a dual RNN that learns patient representations from EHR data. Each RNN encodes only one type of medical code because clinical visits are likely to be missing diagnostic patterns (e.g., only diagnostic patterns but not procedures). Inspired by [17], we use two RNNs to model diagnostic and procedural records separately, projecting the corresponding medical codes into the embedding space via the vector matrix dot product. With EHRs describing patients according to different views, each of which is recorded as $X = [x^1, x^2, \ldots, x^t]$ where t is the total patient visits, and x^t represents records of the tth admission. The EHRs after word embedding will be transferred to the RNN to generate the hidden layer representation based on Eq. (2), generating the patient information p^t:

$$\begin{aligned} h_t^d &= RNN_d\left(x_d^1, x_d^2, \ldots, x_d^t\right) \\ h_t^p &= RNN_p\left(x_p^1, x_p^2, \ldots, x_p^t\right) \end{aligned} \tag{2}$$

Thus, RNN accepts all patient history visits for medical embedding $\{x^*\}(*\leq t)$ generating hidden states $\{h_t^d, h_t^p\}$. The diagnosis and procedure vectors are sent to the attention models. Inspired by [18], we model access-level attention as α according to Eq. (3):

$$\alpha^d = softmax\left(\phi\left(h_1^d\right), \phi\left(h_2^d\right), \ldots, \phi\left(h_t^d\right)\right) \tag{3}$$

where ϕ is a linear transformation function. The records related to diagnostic information are coded by the attention mechanism as β^d, which represents the weight of the different diagnostic records of patients:

$$\beta^d = tanh\left(\psi\left(h_t^d\right)\right), t = 1, 2, \ldots, n \tag{4}$$

where ψ is a linear transformation function. Based on the two attention coefficients generated, we determined the final diagnostic vector o_t^d:

$$o_t^d = \sum_{i=1}^{t} \alpha^d \beta^d \odot x_d^t \tag{5}$$

where $\odot$ denotes element-by-element multiplication. The final representation of the patient procedure cases is also determined using the above method. Finally, we can obtain the patient vector:

$$q_t = w_1 o_t^d + w_2 o_t^p \tag{6}$$

where o_t^d and o_t^p are the final vectors for diagnosis and procedure records, respectively, and w_1 and w_2 are learning parameters. Specifically, we insert the patient history visit vector and corresponding

embedded medication into the history attention module. The attention vector is expected to evolve, ensuring that DGCL always focuses on the most relevant information in the EHR when understanding the patient's condition.

4.3. Clinical records learning module

We have observed that clinicians refer to the patient's diagnosis, procedure, and drug–drug interaction when providing medication combinations for patients. When performing medication recommendation tasks, these medical records can be assigned different weights based on their similarity to the current treatment. Therefore, we developed a two-stage neural network module for the clinical record learning task to address the above problem, as shown in Fig. 2.

The first phase designs a multilayer attention mechanism for patient procedures and diagnosis records. This mechanism effectively aggregates the medical information present in the EMRs. For example, we have diagnostic codes available for some patients but no procedure codes because no procedures have been ordered. Each layer of attention has a different perspective to understand the contribution of the patient's past visit history to the current visit. Once we process different types of medical documents by attention, the model begins the second phase of training, which is to mine all historical cases of patients by distance neural to assist in current medication administration.

4.3.1. Medical history attention

When the model has the patient representation, the corresponding history vector his_{t-1} is expressed as a weighted sum of vectors:

$$\begin{aligned} his &= \sum_{j=1}^{t-1} q_{t-1} + x_m^{t-1} \\ cur &= w_1 o_t^d + w_2 o_t^p \end{aligned} \tag{7}$$

Patients retain their historical memory for each visit. Specifically, the historical attention module is designed for objects with historical cases. Such a strategy can assist the model in identifying historical information related to the current recommendation, which is important because patient medication use is a long and stable process with rich information.

4.3.2. Distance neural network

The primary task of the second stage is to distinguish the link between the current visit and the previous admission records. In a quantitative analysis of the MIMIC-III dataset [19], a patient takes an average of 20 medications and has 5–8 historical cases. When physicians attempt to integrate a patient's past medical history, they must distinguish the impact of past medical history on current medication use, such as some chronic conditions and some acute inflammatory conditions, to develop a list of patient medications.

Therefore, we devise a distance loss function by introducing the Kullback–Leibler (KL) divergence to measure the proximity of historical and current cases [20], retaining the most suitable treatment options . Specifically, to facilitate mutual learning between different information (i.e., past and current cases), we calculate the probability distribution of the distance between consultation records within a fixed time step T [21]. Distance neural networks rethink how historical information can be used from a drug-level perspective by calculating distance loss to determine whether historical drugs are relevant to the current patient visit. The proposed distance loss is shown below:

$$\begin{aligned} \mathcal{L}_D^i &= \frac{1}{n}\sum_{i=1}^{n} D_{KL}\left(\sigma\left(\frac{cur}{T}\right) \| \sigma\left(\frac{his}{T}\right)\right) * T^2 \\ &= \frac{1}{n}\frac{1}{C}\sum_{i=1}^{n}\sum_{s=1}^{C} p_{is}\left(log_{p_{is}} - log_{q_{is}}\right) \times T^2 \end{aligned} \tag{8}$$

where $\sigma(\cdot)$ represents the softmax operation; T is the temperature, where $T = 2$ during the proposed experiments; p_{is} and q_{is} denote the probability that historical case information belongs to the current case predicted by the attention layer and multi-layer perception, respectively.

4.4. Graph contrastive learning

Graph convolutional networks belong to a specific instance of the Laplace smoothing operation. Specifically, graph convolution mixes the features of graph vertices and similar domain nodes. The smoothing operation averages the vertex features within the same agglomerative region; thus, the complexity of the convolved graph structure will be markedly reduced when performing node classification, which is the primary reason why GCNs work.

However, GCNs introduces a potential for over-smoothing in models, where node features can be too monotonous, resulting in nodes from multiple agglomerative regions being too similar to be distinguished. At small scales, blending occurs more quickly.

Specifically, after graph convolution of medical cases, corresponding edges are generated between known combinations of medication use and drug–unknown interactions. However, following graph convolution, the recommended drug features are smoothed and blurred across the DDI network topology, and these features may contain information about drug interaction relationships.

Therefore, it is essential to balance electronic medical record information with DDI network information for drug recommendation. Thus, we develop the graph contrastive learning framework to address these problems. First, we use a two-layer graph convolutional network (GCN) [22], which has a flexible node-embedding architecture, as the graph encoder.

$$GCN(X, A) = \sigma\left(D^{-\frac{1}{2}}\widetilde{A}D^{-\frac{1}{2}}\right) \tag{9}$$

where $X \in \mathbb{R}^{M\times d}$, the medicine adjacency matrix $A \in \mathbb{R}^{M\times M}$, $\widetilde{A} = A+I$, I is the identity matrix and D represents degree matrix of $\widetilde{A}$.

$$\begin{aligned} G_e &= GCN\left(RELU\left(GCN(X_e, A_e)W_e\right), A_e\right) \\ G_d &= GCN\left(RELU\left(GCN(X_e, A_e)W_d\right), A_d\right) \end{aligned} \tag{10}$$

where W_* represents the learnable parameter.

The coded graph is a discrete data structure, and drug nodes tend to be closely associated (such as interactions). Therefore, we design a graph contrast learning algorithm that excavates topological information between nodes concerning the properties of the graph structure. The goal of contrast learning, naturally suited for dealing with drugs requiring discrimination, is to bring similar samples, i.e., the positive ones closer together and the remaining negative ones farther away, thereby maximizing the mutual information between input and positive samples. Specifically, the graph contrast module treats drug combinations with interactions as negative sample pairs, pulling drug combinations for which the DDI exists (negative sample pairs) far by learning the difference between the EHR graph and the DDI graph, reducing the interaction rate of the recommended combinations.

Therefore, we use contrast loss [23] to guide the representation of medical record graphs and drug interaction graphs. The intuition is that medication recommendations and DDIs with serious drug interactions should have different representations (i.e., a graph distance, D, between them). We create negative examples (i.e., mismatches) for training by randomly mixing medication recommendation combinations and DDIs, because the graph contrast encoder requires training with both positive and negative examples of the form (recommended drug, drug interaction graph). For the negative examples, we expect a graph margin (m) between their embeddings. The soft matching contrast loss is defined as follows:

$$D = \|G_e - G_d\|_2 \tag{11}$$

$$\mathcal{L}_{CL} = \mathbb{I}_{positive}\frac{1}{2}(D)^2 + \left(1 - \mathbb{I}_{positive}\right)\frac{1}{2}\{max(0, m - D)\}^2 \tag{12}$$

where the $\mathbb{I}$ match equals 1 if the drug interaction is initially present in the medication recommendation scheme and 0 if the side effect is not in the medicine combination.

Algorithm 1 Training Algorithm for DGCL.

Require: Training set Q, The iterations T_{max}, Trainable parameters, Embedded margin m in Eq.(11), Learning rate η, Initial temperature T in Eq.(8)

Initialize parameters:
Calculate adjacency matrix
for T=1 → T_{max} **do**
  Sample a patient M = $[x_1, x_2, \ldots, x_T]$ from EHR;
  Obtain external knowledge and records from Eq.(9)(10)
  for a visit i=1 to M **do**
    Fetch the $i-th$ visit of patient M, p_t
  Obtain medical embeddings $x_m^t x_p^t$ and x_d^t from Eq.(1)
  Obtain Dual-RNN $\left\{h_t^\alpha, h_t^\beta\right\}$ from Eq.(2)
  Calculate attention weight of medical records from Eq.(3) and(4)
  Generate a patient $M's$ history records H in Eq.(5) ,(6) and(7);
  if Use knowledge **then**
    Based on Eq.(7), propagate over the knowledge graph by the history attention to obtain embedding his_t
    Calculate the distance loss between cases from Eq.(8)
  Calculate contrastive loss $\mathcal{L}_{CL}$ based on Eq.(11)
Accumulate $\mathcal{L}_E$, $\mathcal{L}_D$, $\mathcal{L}_{CL}$ in Eq.(8) ,(11) and(13)
Optimizing the loss in Eq.(13)
return medicine recommendations with the trained model.

4.5. Training and inference

According to the joint training strategy [24], we designed different subtasks in the DGCL : binary cross-entropy loss [25], distance correction loss, and contrast loss. DGCL uses an end-to-end learning mindset and strives to balance recommendation accuracy and drug interaction rates. When the patient's past medical history must be fully considered, we design a distance correction loss to improve the accuracy of model recommendations. In addition, when the interaction rate in the recommended drugs must be controlled, we design the contrast loss function to suppress the DDI in the recommended drugs. We introduce the binary cross-entropy loss when updating the DGCL parameters to minimize the weighted loss.

4.5.1. Multilabel prediction loss (MPL)

Because medication recommendations can be viewed as continuous multilabel predictions, we use a standard multilabel loss function (i.e., binary cross-entropy loss $\mathcal{L}_E$). The binary entropy loss is defined as:

$$\mathcal{L}_E = -\sum_{i=1}^{t}\sum_{k=1}^{N_u}\left(y_i[k]\,log\hat{y}_i[k] + \left(1-y_i[k]\right)log\left(1-\hat{y}_i[k]\right)\right) \tag{13}$$

where N_u is the number of drugs, $y_i[k]$ and $\hat{y}_i[k]$ are the kth predicted and real drug for the ith patient's case.

4.5.2. Objective function

We obtain the full objective function by combining the loss in Eq. (8), (11), and (12). The final loss function of the model is defined as a weighted sum of the different subtask loss functions, as in Eq. (14):

$$\mathcal{L}_{combine} = \alpha_1\mathcal{L}_E + \alpha_2\mathcal{L}_D + \alpha_3\mathcal{L}_{CL} \tag{14}$$

where α_1, α_2, α_3 are hyperparameters to set, we adjust the hyperparameter in Eq. (14) by grid search. Among them, α_1, α_2, α_3 in {0,0.01,0.1,0.25,0.50,0.75,1.00,1.10,1.20}.

A detailed description of DGCL is provided in Algorithm 1.

Table 1
Data statistics.

Attribute	Statistics
No. of patients	6350
No. of diagnoses	1958
No. of procedures	1426
No. of medications	145
Average of visits	2.36
Average of diagnosis	10.51
Average of procedures	3.84
Average of medications	8.80
Medication in DDI knowledge base	448
DDI types in the knowledge base	50

5. Experimental settings

In this section, DGCL is compared with different baseline methods on the real-world clinical dataset MIMIC-III v1.4 [19]. The model is built in PyTorch and trained on an NVIDIA RTX 3080 GPU.

5.1. Dataset

MIMIC-III v1.4 is the richest open source clinical dataset, which includes clinical diagnostic reports of patients admitted to the ICU over an eight-year period since 2001, and reports include laboratory lab results, fluctuations in patient vital characteristics, precautions, etc. The final selected cohort consisted of 6350 patients for 1960 diagnoses, 1432 procedures, and 153 medications.

5.2. Preprocessing

The dataset is coded by the ICD-9 coding system [26], and medication is administered using the NDC (National Drug Code) system [27]. We preprocess the dataset to remove outpatients with only one visit. The first 50 interactions are used for each medication to generate the DDIs adjacency matrix A_d from the Twobedes resource [28] because these interactions include serious primary effects, similar to [7]. Then, the drug co-occurrence adjacency matrix A_e is created. After the above operations are completed, we divide the data into training, validation and testing set in the ratio of $\frac{2}{3}$: $\frac{1}{6}$: $\frac{1}{6}$. The statistics of the postprocessed data are shown in Table 1.

5.3. Hyperparameters settings

In this study, the embedding size d is 64, the batch size during training is 128. We use the Adam optimizer and set the learning rate to 0.002. Also, the dropout rate is set to 0.005 and α_1, α_2, α_3 is set to 1.1, 0.1 and 1.0.

5.4. Baseline methods

We compare DGCL[1] with the following advanced methods:

- **GAMENet** [7] recommends medication treatments via a graphically enhanced memory module with a dynamic network based on medication use and DDIs.
- **SafeDrug** [29] proposes a DDI-controlled medication recommendation model to exploit the molecular structure of the medication and the DDI model. SafeDrug has designed a novel knowledge propagation neural network and a local molecular structure to assist recommendations.
- **DMNC** [10] proposes a model for medication combination prediction in memory-enhanced neural networks based on a microscopic neural computer.

[1] https://github.com/Benjamin-cell/DGCL

Table 2
Performance comparison on MIMIC-III(the base DDI rate is 0.0808).

Methods	DDI rate	△ DDI rate %	Jaccard	F1-score	PRAUC	AvgDrugs
LEAP	0.0731 ± 0.0008	−9.53%	0.4521 ± 0.0024	0.6138 ± 0.0026	0.6549 ± 0.0033	18.7138 ± 0.0666
DMNC	0.0842 ± 0.0011	+4.21%	0.4864 ± 0.0025	0.6529 ± 0.0030	0.7580 ± 0.0039	20.0000 ± 0.0000
GAMENet	0.0864 ± 0.0006	+6.93%	0.5067 ± 0.0025	0.6626 ± 0.0025	0.7631 ± 0.0030	27.2145 ± 0.1141
AMANet	0.0883 ± 0.0015	+9.28%	0.5152 ± 0.0034	0.6752 ± 0.0028	0.7672 ± 0.0024	14.3168 ± 0.3115
MRSC	–	–	0.5047 ± 0.0034	0.6618 ± 0.0031	0.7705 ± 0.0021	–
SARMR	0.0627 ± 0.0011	−22.40%	0.5039 ± 0.0023	0.6608 ± 0.0024	0.7688 ± 0.0032	21.3164 ± 0.1329
MICRON	0.0641 ± 0.0007	−20.67%	0.5100 ± 0.0033	0.6654 ± 0.0031	0.7687 ± 0.0026	17.9267 ± 0.2172
SafeDrug	0.0589 ± 0.0005	−27.10%	0.5213 ± 0.0030	0.6768 ± 0.0027	0.7647 ± 0.0025	19.9178 ± 0.1604
PREMIER	0.0750 ± 0.0000	−7.18%	0.5270 ± 0.0000	0.6810 ± 0.0000	0.7800 ± 0.0000	22.6503 ± 0.0000
COGNet	0.0852 ± 0.0005	+5.45%	0.5336 ± 0.0011	0.6869 ± 0.0010	0.7739 ± 0.0009	28.0903 ± 0.0950
4SDrug	0.0600 ± 0.0004	−25.74%	0.5041 ± 0.0016	0.6581 ± 0.0016	0.7640 ± 0.0013	17.5040 ± 0.0533
DGCL	**0.0587 ± 0.0003**	**−27.35%**	**0.5342 ± 0.0016**	**0.6874 ± 0.0021**	**0.7824 ± 0.0009**	**28.6253 ± 0.0827**

– Represents that the experimental results are not given in the original study.

- **Leap** [6] performs recommendations according to individual instances, and analyzes individual patient cases to recommend personalized drug.
- **MRSC** [3] is a selective coverage multihop reading model. MRSC covers EHR long-term attention calculations , by skipping to ensure that important histories are considered.
- **MICRON** [30] recommends drugs based on medication changes and the last patient visit.
- **AMANet** [17] is conducted on a two-view sequence. Specifically, the model learns the patient's EHR from two perspectives.
- **SARMR** [4] uses the self-supervised adversarial regularization medication recommendation model, which obtains target distributions that are associated with drug combinations from the original patient records.
- **COGNet** [31] is a recommendation model based on the transformer architecture, which mainly assists the current recommendation by copying historical drugs.
- **4SDrug** [32] is a small and safe drug recommendation model based on symptoms.
- **PREMIER** [33] uses graph convolutional networks to realize personalized medicine recommendations.

5.5. Evaluation metrics

Following the previous medicine recommendation works [7,29], we use the following metrics: Jaccard similarity (Jaccard), precise recall AUC (PRAUC), F1 score and average number of recommended drugs(AvgDrugs) as evaluation metrics. Inspired by [29], we use the DDI rate to measure the safety of the recommended drugs with the following formula:

$$DDI\,Rate = \frac{\sum_{k}^{N}\sum_{t}^{T_k}\sum_{i,j}\left|\left\{\left(c_i, c_j \in \hat{Y}_t^k\right)\right\}\right|\left(c_i, c_j \in \varepsilon_d\right)}{\sum_{k}^{N}\sum_{t}^{T_k}\sum_{i,j} 1} \tag{15}$$

where the set counts each medication pair (c_i, c_j) in recommendation set $\hat{Y}$ if the pair belongs to the edge set ε_d; N is the test dataset; and T_k is the number of visits of the kth patient. After recommendation models calculate the DDI rate, the percentage of DDI change relative to the DDI rate of the EHR test data set is defined as △ DDI Rate %:

$$\triangle\, DDI\,Rate\% = \frac{DDI\,Rate - DDI\,Rate\,(EHR)}{DDI\,Rate\,(EHR)} \tag{16}$$

6. Experimental results

6.1. Overall performance

In this section, we compare DGCL with existing works on medicine recommendation, and the result is reported in Table 2. By analyzing the experimental results, we draw some conclusions:

- Some attention and graph convolution neural networks (e.g., AMANet, GAMENet, ARMR) outperform other models in terms of recommendation accuracy (Jaccard, F1), indicating that such models can manage the drug recommendation task well. DDI suppression (SafeDrug, SARMR) keeps the DDI rate within a relatively reasonable range while ensuring recommendation accuracy because SafeDrug and SARMR have designed their neural networks to manage recommendations and use regularization methods to control the DDI in the recommended drugs. Existing neural networks achieve good recommendation accuracies if only the drug recommendation task is considered; for example, AMANet achieves the best results on the F1 value and is competitive in other metrics. However, these models fail to control the DDI effectively, and the recommended drug combinations have many interactions, which defeats the purpose of this study's goal to recommend safe medicines. Recommendation models should achieve high recommendation accuracy at low DDI rates to provide safer drug combinations in a reasonable state.
- The proposed DGCL outperformed nearly all other models based on the evaluation metrics. In particular, DGCL achieves the lowest DDI rate while still maintaining a high recommendation accuracy, demonstrating the validity of the proposed model. DGCL's advantage over SafeDrug is evident, with gains of 2.47% and 2.31% in both Jaccard and PRAUC. In addition, we can draw some conclusions from the average number of medications recommended. We observe that LEAP and AMANet recommend less number of drugs. Both models fail to mine historical records related to current visits, so the number of recommended drugs is limited.
- Regarding recommended safety, 4SDrug is designed with knowledge-based and data-driven penalties to ensure safer drug recommendation, controlling DDIs at a low level. Compared with 4SDrug, DGCL further reduces the DDI rate and achieves a lift of 5.97% and 4.45% on Jaccard and F1 values. It is because the clinical record learning module focuses on mining the patient's health history to assist recommendations, calculating how similar the records are in different periods, and looking for pertinent medical records. In terms of recommended accuracy, both COGNet and PREMIER achieve high recommended accuracy. However, COGNet does not consider the recommended DDI rate of drugs, resulting in a poor safety profile of the drugs, and PREMIER also failed to suppress the DDI rate further. In contrast, DGCL improves the recommended accuracy and outperforms COGNet and PREMIER in terms of DDI control, reducing the DDI rate by 27.35%.
- DGCL recommended the highest number of drugs, suggesting that the model achieved a more comprehensive recommendation outcome despite a lower drug interaction rate, striking a better balance between safety and accuracy of recommendations.

Table 3
Performance comparison of ablation study.

Methods	F1	DDI	Jaccard	PRAUC	AvgDrugs
No-HM	0.6533 ± 0.0013	0.0653 ± 0.0003	0.4871 ± 0.0015	0.7516 ± 0.0014	26.5377 ± 0.0842
No-GDN	0.6485 ± 0.0015	0.0648 ± 0.0005	0.4793 ± 0.0013	0.7431 ± 0.0011	28.4781 ± 0.0592
No-KL Loss	0.6606 ± 0.0011	0.0683 ± 0.0002	0.4984 ± 0.0021	0.7613 ± 0.0019	29.5274 ± 0.0831
No-GCL	0.6732 ± 0.0016	0.0886 ± 0.0004	0.5218 ± 0.0014	0.7702 ± 0.0014	28.6823 ± 0.0724
No-DDI	0.6696 ± 0.0013	0.0821 ± 0.0003	0.5202 ± 0.0011	0.7694 ± 0.0015	29.3782 ± 0.0705
DGCL	**0.6874 ± 0.0021**	**0.0587 ± 0.0003**	**0.5342 ± 0.0016**	**0.7824 ± 0.0009**	**28.6253 ± 0.0827**

Table 4
T-test analysis results.

Metrics	Sample Size	T	*P*-value	Mean
PRAUC	10	−48.428	3.422×10^{-12}	0.782(DGCL) 0.683(COGNet)
F1	10	−8.773	4.605×10^{-7}	0.691(DGCL) 0.674(COGNet)
Jaccard	10	−20.921	2.154×10^{-11}	0.537(DGCL) 0.522(COGNet)

- We reproduce DGCL and COGNet under the same experimental setting.

6.2. Ablation study

In this section, five variant versions of DGCL are developed to investigate the contribution of each component of the model, and we compare these five variants experimentally with the DGCL model on MIMIC-III. Details of these variants are described below:

- **No Graph Contrastive Learning (No-GCL)**: DGCL takes the EHR and DDI information encoded by the graph convolutional network for contrastive learning. No-GCL does not perform contrastive learning but simply concats the two pieces of information together.
- **No Graph Distance Neural Networks (No-GDN)**: DGCL corrects the distribution of historical and current cases by minimizing the KL divergence of these two components. No-GDN does not calculate the distance between these cases but rather uses the current patient's diagnostic and procedure records for medication recommendations.
- **No-KL loss**: To further investigate the contribution of the graph distance module, we retain patient history information and remove the loss of distance between different clinical records.
- **No DDI Knowledge (No-DDI)**: No-DDI does not use drug interaction matrix information but simply encodes electronic medical record information.
- **No History Memory Module (No-HM)**: No-HM does not consider a patient's hospital history when recommending medication to a patient.

We show the results of these five variants in Table 3. Each component consistently makes a different contribution to the dataset. Removing the clinical records learning module resulted in the largest decrease in F1. The clinical learning module maximized the base model and improved the accuracy of the recommended drugs. To further investigate the internal mechanism of the module, we disassemble the module into two components(No-HM and No-KL) to analyze the impact of the different components on the model performance. A decrease in F1 is shown when the history and KL module are removed; this phenomenon is interesting because these two components have different principles but both consider the diagnosis and procedure files. Therefore, we will further investigate their connection in more detail during the recommendation task in the case study. In addition, external DDI knowledge and contrastive learning frameworks effectively suppress DDI. The DDI also reaches a maximum in the absence of the contrastive learning framework. Concurrently, the DDI rate still reaches a high level when the DDI knowledge graphs are not used. Thus, the DDI knowledge graphs and the contrastive framework are shown to be effective. The DDI is also lower on three components(No-HM, No-KL loss, and No-GDN) due to poorer accuracy of recommendations, fewer drug combinations, and therefore a lower DDI rate, which is a reasonable situation that is not due to the model controlling the DDI.

6.3. Statistical significance test

In this section, we conduct a statistical significance test to evaluate the performance improvement of DGCL. Specifically, we conduct an independent sample t-test [34] on different evaluation indicators to analyze whether significant differences exist between DGCL and baseline performance. We compare the strongest baseline COGNet and DGCL and randomly select ten groups of F1, AUC, and Jaccard values on the test set. Table 4 shows the significance test results. For the independent sample t-test, the experimental results are generally analyzed in two steps. First, analyze whether the evaluation indicators of different models are significant, generally measured by p-value. When the p-value is less than 0.05 or 0.01, it can be judged that there is a significant difference between the two data groups. Secondly, when the two data groups show significant differences, it is necessary to compare the mean value and describe the specific differences.

It can be seen from the Table 4 that the test samples of DGCL and COGNet show significant differences ($p < 0.01$) for all three types of evaluation indicators, which means that the samples of the two models have significant differences for indicators. Taking the difference analysis of F1 as an example, the test samples show a significant 0.01 level for F1 ($t = -8.773$, $p - value = 4.605 \times 10^{-7}$). Moreover, the mean value of COGNet is 0.674, significantly lower than that of DGCL (0.691). It can be concluded that the test data of COGNet and DGCL show significant differences for all three types of evaluation indicators.

6.4. Analysis of DDI rate

In this section, we evaluate the controllability of DDI and show that the DDI rate can be controlled by the proposed model. We follow the same criterion [10] with a valid DDI rate of 0.0808 for MIMIC-III. Therefore, the medicine recommendation should optimize the final recommendations by processing EHRs and drug–drug external knowledge, and the DDI rate should be lower than the human experts. In this section, we design two model variants to test (as shown in Fig. 3).

- **DDI(with GCL)**: This method is a simplified variant of the original DGCL model without changes and serves as the baseline for the experiments.
- **DDI(without GCL)**: This method is a simplified variant of the original model, removing the graph contrastive learning module.
- **DDI(with EHR Graph)**: This method retains the graph contrastive learning module, but the comparison subgraphs are no longer DDI external knowledge graphs. Also, all patient cases are used as comparison subgraphs. This variant hopes to learn from the existing medication cases and reduce the DDI rate.

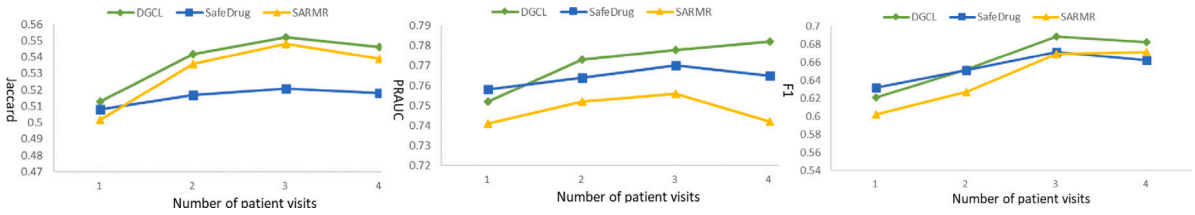


Fig. 3. Analysis of historical information.

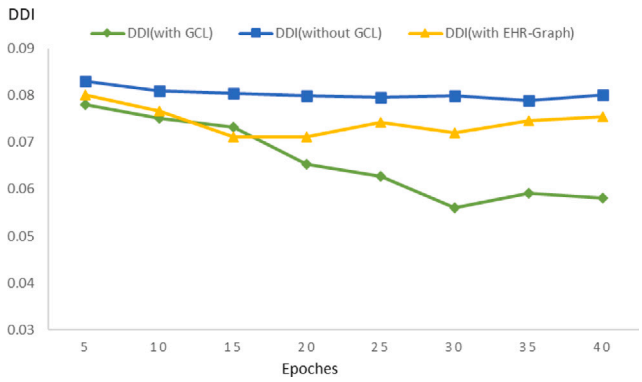


Fig. 4. Analysis of DDI rate.

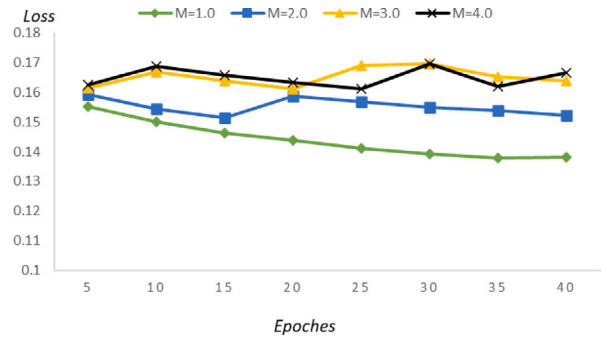


Fig. 5. Analysis of parameter sensitivity.

As shown in Fig. 3, DGCL controls DDI better than the other two variants. DGCL achieves the lowest DDI rate after approximately 30 rounds, demonstrating the proposed method's effectiveness. In addition, the DDI rate without using the graph contrastive framework remains relatively unchanged and high. Finally, the variant that uses patient cases for comparison converges quickly within the first 15 rounds and performs marginally better than DGCL at times, which is reasonable because the EHR graph is simpler and involves fewer drug entities than the DDI knowledge graph. Therefore, the variant can control interactions in the early training stage and reaches the optimal value of approximately 15 rounds. However, this approach does not guide the model to reduce the DDI rate later, and the DDI is always maintained in a more stable interval.

6.5. Analysis of historical information

To analyze whether the DGCL captures patient historical information, we explore the effect of the number of visits on the recommended outcomes. Because most patients visited the hospital no more than four times, we screen four visits for each patient in the dataset. We also use two efficient baselines SafeDrug and SARMR for analysis. Fig. 4 shows the experimental results. DGCL achieves better performance as the number of visits increases, while the performance of the SafeDrug remains almost flat, and SARMR exhibits a downward trend. These results may occur because DGCL uses a weight update mechanism based on historical attention, followed by an additional distance neural network that simulates the connections between different patient information. The two-stage training mechanism allows for an efficient merging of information from past visits. In addition, SARMR updates past drug combinations to form new recommendations, which limits the use of historical information.

6.6. Parameter sensitivity

In this section, we evaluate how the hyperparameters affect the performance of DGCL and the smoothness of the graph convolution. Specifically, we evaluate the impact of the hyperparameter M in the graph contrastive framework, measuring the convergence of the model. In addition, this section tests the effect of the graph layers.

6.6.1. Convergence analysis

In this section, we measure the convergence of the experiments by plotting the loss curves, reporting the loss development at different M values. Each experiment is repeated five times, and the results are shown in Fig. 5. The curve decreases more gently when M equals 1, which indicates a better convergence of the graph contrastive framework. In addition, we also find that the loss curve has local jitter in rounds 15–20 when M equals 2. The contrast subgraph achieves a local optimum, and thus a new round of convergence is performed in the subsequent training. When continuing to increase M, the loss curve repeatedly oscillates, indicating that the model converges less well at this point. This result is reasonable because sampling negative samples with a marginal M that is too large leads to the introduction of sampling bias in the subgraphs, resulting in larger losses and difficulty in convergence.

6.6.2. Influence of the number of layers

To investigate the number of layers in the graph convolutional network, we divide the range of layers by 1,2,3,4,5 and record the average of the results. According to the results shown in Fig. 6, the proposed model achieves the best performance with two to three-layer settings. When the number of layers is increased, performance decreases due to oversmoothing. In addition, significant fluctuations are observed in the DDI, which decreases before two layers, and the DDI continues to decrease after increasing the number of layers, due to a decrease in the recommended accuracy, not an active suppression of the DDI rate.

6.7. Robustness analysis

To explore the robustness of recommendation models, we investigate the DDI rate on different DDI types. We perform experiments using three baselines, GAMENet, SafeDrug and COGNet, designed with different strategies to control DDI rate.

Table 5 shows the results. We observe some interesting phenomena: as the DDI types increase, GAMENet, SafeDrug and COGNet show degradation in suppressing the DDI rate. GAMENet reduces the DDI rate by 2.22% when the top 40 DDI types are used, but GAMENet fails when more interaction types are considered. As the number of

Table 5
Performance comparison of DGCL with other methods in different DDI types.

DDI type	DDI rate	GAMENet	SafeDrug	COGNet	DGCL
40	0.0808 ± 0.0003	0.0793 ± 0.0006	0.0589 ± 0.0005	0.0852 ± 0.0005	0.0512 ± 0.0003
60	0.1973 ± 0.0005	0.2106 ± 0.0005	0.2117 ± 0.0013	0.2681 ± 0.0011	0.1887 ± 0.0592
80	0.2964 ± 0.0011	0.3183 ± 0.0002	0.3095 ± 0.0021	0.3451 ± 0.0019	0.2676 ± 0.0831
100	0.3971 ± 0.0006	0.4272 ± 0.0004	0.4106 ± 0.0014	0.4536 ± 0.0014	0.3757 ± 0.0724

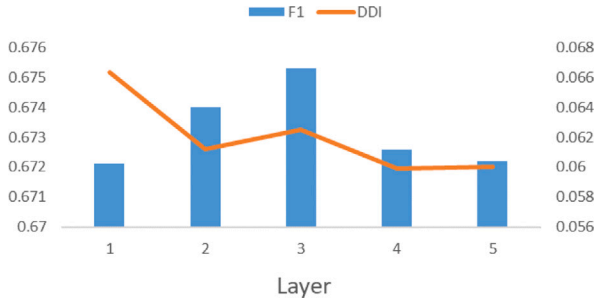


Fig. 6. Analysis of the graph layer.

DDI types increases from 60 to 100, GAMENet's DDI rates increase by 6.74%, 7.39% and 7.58% respectively compared to the original data. In addition, COGNet fails to reduce DDI in any cases. It is because COGNet is based on the replication mechanism that learns the patient's historical medication to finish recommendations. The DDI rate of medicines will continue to accrue as historical medicines are reused, resulting in the increasing DDI rate.

In addition, SafeDrug is the better performing baseline, suggesting that molecular knowledge can suppress interaction rates in drug combinations. However, when the DDI types increase, SafeDrug's DDI rates increase by 7.30%, 4.42% and 3.40% respectively, which suggests that relying on knowledge to suppress DDI rates is insufficient. The results demonstrate the robustness of DGCL as it is the only model to control DDI reduction when DDI types are increasing.

6.8. Case study

In this section, we consider a specific patient case to demonstrate the model's validity. We have removed personal information. Figs. 7 and 8 show the similarity between different types of medical records. Specifically, we use the horizontal axis to represent the patient's procedure codes and the vertical axis to represent the diagnosis codes, and the squares in the graph indicate the degree of similarity between records, with strongly correlated records represented in red and less correlated ones in white.

The patient had a history of emphysema (not on oxygen at home) and what the attending physician believed to be an episode of chronic obstructive pulmonary disease (COPD), resulting in persistent respiratory failure and dyspnea. The patient was intubated to improve oxygenation, ventilation, and respiratory capacity. She presented again on the medical ward with chest pain, but no electrocardiogram changes were noted. The patient was hypothyroid and required continued outpatient care. She was being treated for this symptom because she was depressed for most of her hospital stay and did appear to be clinically depressed.

Therefore, the patient was diagnosed with a malignant liver tumor and drug-induced psychosis and subsequently underwent a liver transplant. During the patient's follow-up treatment, she suffered systemic shock from a viral infection, and such symptoms are a common cause of death in intensive care units. Because shock often causes multiple organ damage, early diagnosis and treatment are necessary to reduce morbidity and mortality. In this case, we focused on observing all the recommended medications. Fig. 7 shows the distribution of patient

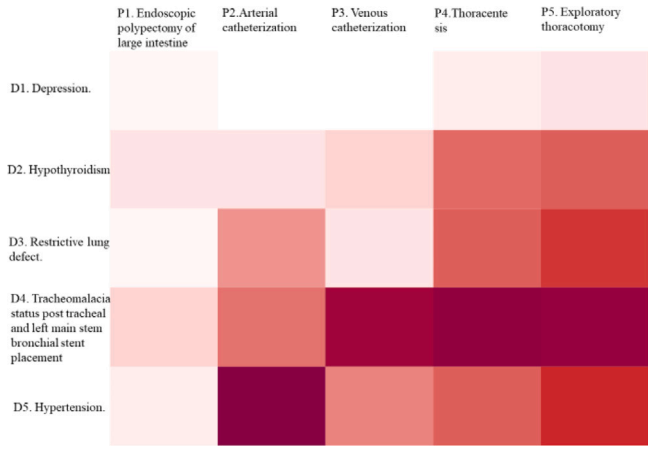


Fig. 7. DGCL with GDN module.

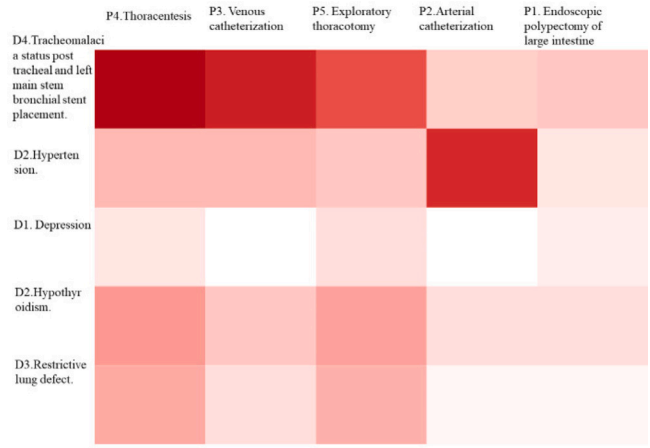


Fig. 8. DGCL without GDN module.

medication. We observe that diagnosis accounted for nearly 50% of recommended medications, surgery accounted for nearly 40%, and past medications accounted for the remaining 12%.

In addition, we observe the clinical record learning module(GDN) improves medical code representation. As shown in Fig. 7, procedure codes P1,P2,P3 and P5, diagnostic codes D1 and D4 are all related to respiratory disease. Unlike the other treatment operations in this study, procedure code P4 is a polyp removal procedure for the large intestine. In addition, hypertension (D2) and depression (D5) remained untreated and should not be associated with any procedure code.

Moreover, the patient with bronchiectasis (D4) is treated by a series of procedure codes. The procedure codes (P2,P3,P4,P5) are all related to the bronchial treatment, corresponding to the darker color of the matrix in Fig. 7, which also shows that Fig. 7 is consistent with the medical facts described above. In stark contrast, when the GDN is removed, the DGCL fits poorly. We observe that the matrix color difference in Fig. 8 is not significant, which indicates that the DGCL(without GDN model) cannot combine the associated diagnosis and procedure codes.

DGCL tends to assign weights equally to the consultation information, resulting in a poor representation of medical codes.

7. Limitation

This study balances the accuracy of drug recommendations with DDI rates, achieving lower drug interactions and better recommendations. However, the distinction between drug interactions is not sufficiently detailed, and future work should distinguish between drug interactions with synergistic effects, interactions with inhibitory effects, and drug interactions with adverse reactions to assist in a more rational recommendation task.

8. Conclusion

In this paper, we present the model DGCL, which aims to generate effective and safe medication combinations via graph neural networks. Specifically, the proposed model extracts and encodes rich information on drug interactions to reduce interactions in medications. Then, we design a two-stage neural network for clinical record learning to provide case information. In addition, the DGCL mitigates interactions between recommended medications through graph contrastive learning. Finally, extensive experiments on the real EHR dataset demonstrate that DGCL outperforms existing networks on nearly all metrics and achieves the lowest DDI rate and the highest number of recommended medications. A potentially valuable pathway for future research is to predict potential drug interactions through external knowledge and recommend medication combinations with a higher safety profile based on the predictions.

CRediT authorship contribution statement

Xingwang Li: Designed the method and experiments, Performed the experiments and analyzed the results, Wrote the manuscript. **Yijia Zhang:** Designed the method and experiments, Provided suggestions and feedback. **Xiaobo Li:** Performed the experiments and analyzed the results. **Hao Wei:** Wrote the manuscript. **Mingyu Lu:** Provided suggestions and feedback.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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